LARISA Y.C. LOKE

larisaycl@gmail.com ◇ larisaycl.github.io

SUMMARY

Roboticist building assistive systems around how real people actually use them, with expertise in human input signal interpretation, ML-based intent inference, and minimal-assumption shared control, implemented in ROS 2, Python, and C++ on real hardware and validated in longitudinal studies with motor-impaired users. Currently a Postdoctoral Researcher at Northwestern University and Shirley Ryan AbilityLab, developing assisted driving paradigms for powered wheelchairs.

EDUCATION

PhD in Mechanical Engineering, Northwestern University, Evanston, IL — Sep 2020 - Jul 2026

- **Thesis:** Assistive Robotics for the Individual: Challenging Assumptions from Signal Interpretation to Shared Control
- **Research Areas:** Human-Robot Interaction (HRI), machine learning, human intent inference, personalization, accessible control systems
- **Awards:** Walter P. Murphy Fellowship (2020-2021)
- **Served as Reviewer:** HRI, THRI, WAFR, ICORR, T-BME, T-ASE

BEng in Mechanical Engineering, Nanyang Technological University (NTU), Singapore — Aug 2016 - Jul 2020

- Honours with Highest Distinction, Robotics and Mechatronics Specialization
- **Awards:** Nanyang Scholarship, CN Yang Scholars Programme, Mechanical Engineering Dean's List

WORK EXPERIENCE

Research Intern, General Motors Research & Development, Warren, MI — Jun - Sep 2025

- Led design and implementation of an accessible in-vehicle interaction system using ROS 2, integrating AAC devices and USB/HID inputs.
- Designed a framework and built extendable software architecture for arbitrating control between human and automated agents.
- Designed a safety-driven state machine governing the full in-vehicle interaction sequence for reliable, predictable behavior.

RESEARCH EXPERIENCE

Postdoctoral Researcher / PhD Candidate, Northwestern University & Shirley Ryan AbilityLab, Chicago, IL
Advisor: Dr Brenna D. Argall — Sep 2020 - Jul 2026 (PhD); Jul 2026 - Present (Postdoc)

Key Responsibilities

- Design and develop algorithms for intelligent robot control, teleoperation, and human-robot interaction.
- Design and conduct virtual and physical HRI experiments (screen-based tasks and real-world robot ADL tasks) with motor-impaired populations to evaluate methods and algorithms.
- Lead human subjects research operations: IRB applications, study protocol design, ethical compliance, participant recruitment/scheduling, conducting study sessions, and secure data de-identification and management.
- Build data collection, processing, and analysis pipelines; analyze results and communicate findings through publications.
- Collaborate with lab members, clinicians, and industry partners on robot learning, user training, user-defined control maps, and robot navigation projects.
- Mentor junior PhD students, master's students and undergraduates on research projects.

Main Research Projects

- **Intelligent Powered Wheelchair** *(Current work, Published in RSS 2026)*
  - Developed a reactive shared-control scheme for local guidance around obstacles during powered wheelchair driving.
  - Built a modular ROS 2 (Python/C++) shared-control and teleoperation stack for a commercial powered wheelchair with LUCI assistive driving technology, handling real-time human control input interfacing, input source arbitration, sensor stream processing and fusion, odometry, and navigation.
  - Designed and conducted a large-scale, longitudinal quantitative user study (over 90 sessions) evaluating usability and effectiveness across diverse user profiles and control interface modalities in indoor and outdoor mobility tasks.
  - Reviewed existing and developed robotic wheelchair human-robot interaction evaluation metrics spanning user, clinical, and engineering perspectives.
  - Co-developed the LUCI Dev Kit, an open research-and-development platform that this work is built on, turning a research prototype into a reusable tool for the wheelchair-robotics research community *(manuscript under review)*.
- **Interface-aware Assistive Control Framework** *(Published in ISER 2023, RSS 2025, IJRR 2025)*
  - Developed algorithms for real-time user-intent estimation using time-series ML models (LSTM/GRU), enabling people using a low-dimensional sip/puff interface to control a high-DoF robot arm (Kinova Jaco) for activities of daily living.

  - Implemented end-to-end data collection → model-training → rollout pipeline to evaluate usability and performance of assisted teleoperation frameworks in two human-subjects studies (3 and 8 participants with spinal cord injury), and iterated the modeling approach based on evaluation findings.
  - Deployed the framework as both an assisted control and training interface, enabling users to provide high-quality robot demonstration data for downstream robot-learning research.
- **Eye Gaze Tracking and Mental Workload Estimation for Wearable Interfaces** *(Published in ICORR 2023, Wearable Technologies 2025)*
  - Designed and implemented a data collection and modeling framework for eye-gaze signals (using the Tobii Pro Glasses 3), studying 11 participants with ALS and SCI to develop methods distinguishing intentional from reactionary gaze, enabling its use in richer control schemes.
  - Analyzed gaze movement, pupillometry, and heart rate data; identified opportunities for customization of gaze-based control interfaces, and demonstrated the feasibility of using gaze signals to estimate cognitive workload during concurrent eye-based control.

Visiting Student Researcher, Robotics Institute, Carnegie Mellon University, Pittsburgh, PA Jun - Dec 2019
Supervisors: Dr John M. Dolan, Dr Christoph Mertz

- Trained CNN-based computer vision models (Faster R-CNN, PyTorch) for rare-object detection, including a synthetic data generation pipeline (image compositing, augmentation) to address limited training data for underrepresented classes.

PUBLICATIONS

- **Larisa Y.C. Loke***, Joel X. Goh*, Jered Dean, Brenna Argall. "A Robotic Wheelchair Research & Development Platform: LUCI Dev Kit." *Under review. *Equal contribution.*
- **Larisa Y.C. Loke**, Joel X. Goh, Kyle S. P. Puckett, Michelle H. Zhang, Nicole L. Zaino, Brenna D. Argall. "A Minimal, Deterministic Approach to Shared Control for Safe Powered Wheelchair Driving." *In Proceedings of the Robotics: Science and Systems (RSS) conference, 2026.*
- Kyle S. P. Puckett*, **Larisa Y.C. Loke***, Nicole L. Zaino, Brenna D. Argall. "A Survey of the Evaluation of Robotic Wheelchairs." *Under review. *Equal contribution.*
- Demiana R. Barsoum, Mahdieh Nejati Javaremi, **Larisa Y.C. Loke**, and Brenna Argall. "Interface-Aware Trajectory Reconstruction for Robot Learning." *In Proceedings of the ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2026.*
- **Larisa Y.C. Loke** and Brenna D. Argall. "Interface-Level Intent Inference for Environment-Agnostic Robot Teleoperation Assistance." *In Proceedings of the Robotics: Science and Systems (RSS) conference, 2025.*
- Mahdieh Nejati Javaremi, **Larisa Y.C. Loke**, Brenna D. Argall. "Task-agnostic and interface-aware handling of unintended interface operation for robot arm teleoperation: Insights and case study evaluations by persons with spinal cord injury." *The International Journal of Robotics Research (IJRR) Special Issue: ISER 2023, 2025.*
- Demiana R. Barsoum, Michelle H. Zhang, **Larisa Y.C. Loke**, and Brenna Argall. "Curating Interface Maps for Robot Teleoperation." *In Proceedings of the International Conference on Rehabilitation Robotics (ICORR), 2025.*
- **Larisa Y.C. Loke**, Demiana R. Barsoum, Todd D. Murphey, and Brenna D. Argall. "Characterizing Eye Gaze and Mental Workload for Assistive Device Control." *Wearable Technologies 6, e(13). Cambridge University Press, 2025.*
- Mahdieh Nejati Javaremi, **Larisa Y.C. Loke**, Brenna D. Argall. "Interface-Aware Assistance for 7-DoF Robot Arm Teleoperation: Case Studies on Feasibility." *In Proceedings of the International Symposium on Experimental Robotics (ISER), 2023.*
- **Larisa Y.C. Loke**, Demiana R. Barsoum, Todd D. Murphey, and Brenna D. Argall. "Characterizing Eye Gaze for Assistive Device Control." *In Proceedings of the International Conference on Rehabilitation Robotics (ICORR), 2023.*
- Andrew Thompson*, **Larisa Y.C. Loke***, and Brenna D. Argall. "Control Interface Remapping for Bias-aware Assistive Teleoperation." *In Proceedings of the International Conference on Rehabilitation Robotics (ICORR), 2022. *Equal contribution.*

POSTER PRESENTATIONS & WORKSHOP TALKS

- **Larisa Y.C. Loke**, Joel Goh, Kyle Puckett, Nicole L. Zaino, Brenna D. Argall. "REACT + LUCI: Reactive Driver Assistance for Powered Wheelchairs." Late Breaking Abstract Poster presented at RehabWeek 2025.
- Andrew Thompson, Michele Lee, **Larisa Y.C. Loke**, Brian Martinez, Mahdieh Nejati Javaremi, and Brenna D. Argall. "Identifying Accessibility Barriers to Robotics Research." Presented at Robotics: Science and Systems (RSS) 2023 Workshop — Lowering Barriers for Robotics Research. *Randomized author order.*

WORKSHOP ORGANIZATION

Human-Robot Contact and Manipulation Workshop, at the 2025 Robotics: Science and Systems Conference

- Co-Organized with Ethan K. Gordon, Tianyu Li, Andrew Thompson, Vinitha Ranganeni, Nadia Figueroa, Tapomayukh Bhattacharjee.

- Responsibilities included website management, coordinating workshop abstract submissions and reviews, and assisting with in-person workshop facilitation.

VOLUNTEER ACTIVITIES

Student Volunteer, RehabWeek 2025, Chicago, IL May 2025

- Facilitating registration and check-in to workshop and conference sessions.
- Conducted Shirley Ryan AbilityLab tours and demoed lab projects.

Lab Demo Volunteer

- Demo and present robotic platforms to children.
- Robot Block Party, National Robotics Week, Museum of Science and Industry, Chicago, IL Apr 2022, 2023, 2025
- National Biomechanics Day, Shirley Ryan AbilityLab, Chicago, IL Apr 2023, 2024

TEACHING EXPERIENCE

Teaching Assistant, Northwestern University

- Hosted lab sessions and office hours to assist students with code and hardware. Graded student assignments and final exams, and provided written feedback on reports. Gave lectures.
- Courses:
  - CS/ME 469 Machine Learning and Artificial Intelligence for Robotics Sep 2023 - Dec 2023
  - CS/ME 301 Introduction To Robotics Laboratory Jan 2022 - Mar 2022, Jan 2023 - Mar 2023
  - ME 449 Robotic Manipulation Sep 2021 - Dec 2021

SKILLS

Programming Languages	Python, C++, Arduino, MATLAB, C# (Unity)
Tools	Robot Operating System (ROS & ROS 2) and associated packages, PyTorch, scikit-learn, Pandas, Docker, Linux, Git/GitHub, LaTeX
Robot Platforms	Kinova JACO Gen2, Kinova MICO, UFactory xArm 7, LUCI smart wheelchair
Control Interfaces	Eye gaze tracking, Sip/Puff, Joysticks, Head Array, Head Mouse
Languages	English (Native), Mandarin Chinese (Proficient)